

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460668.9452 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.013
Transit depth (Rp/Rs)^2	1.12 +/- 0.28 [%]
Orbital Inclination (inc)	82.35 +/- 2.39
Airmass coefficient 1 (a1)	1.0336 +/- 0.0073
Airmass coefficient 2 (a2)	-0.0329 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	#2 - [296, 301]
Optimal Aperture	4.19
Optimal Annulus	5.93
Transit Duration (day)	0.1173 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.106 ± 0.013 vs 0.1178 (deviation 0.91σ)
Duration fit vs published	0.1173 d vs 0.125 d (deviation 1.79σ)
Mid-time O-C	-3.9 min
Depth SNR	8.2
Residual scatter	0.7 %

Light curve

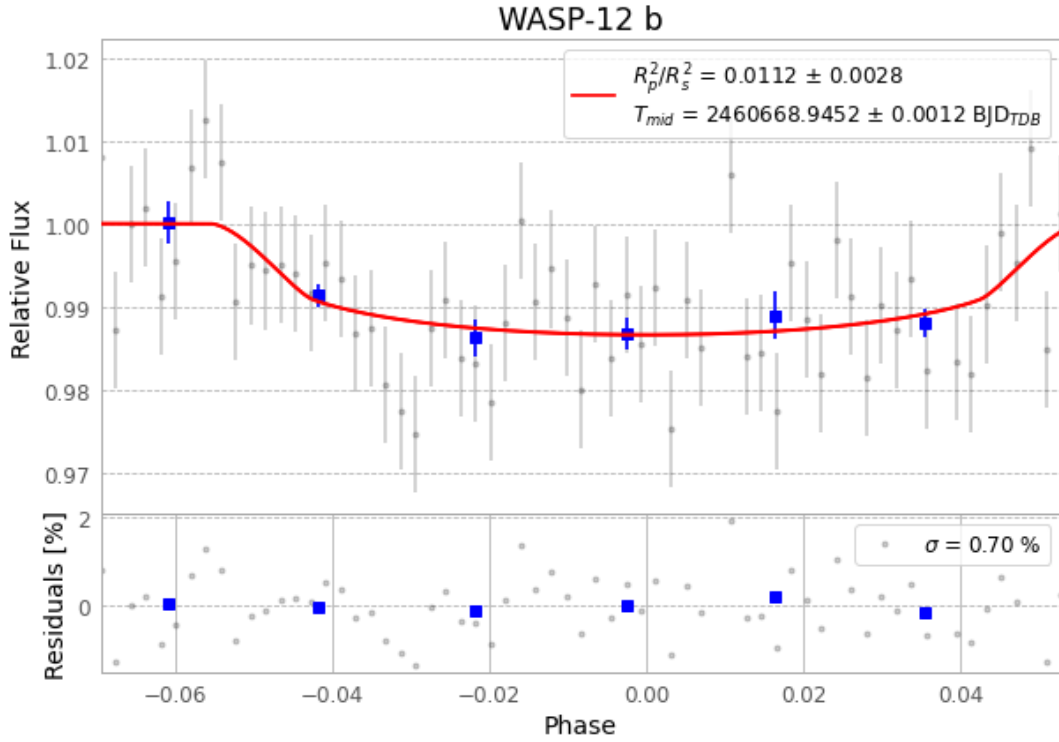


Figure 1 — FinalLightCurve_WASP-12 b_2024-12-24.png (SHA-256 f301e9d3c8fb756a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8a3ea2e434d7e118...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

65 FITS frames (43 MB), WASP-12241224084216.FITS → WASP-12241224115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241224.log (12359d8c592e...), FinalLightCurve_WASP-12 b_2024-12-24.png (f301e9d3c8fb...), FinalLightCurve_WASP-12 b_2024-12-24.csv (10040ddd4a0e...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 01:18Z.